Daniel McKirby

Boston, MA | U.S. Person / ITAR Eligible
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EDUCATION

Northeastern University Boston, MA
B.S. in Mechanical Engineering, Minor in Aerospace *Candidate, Dec 2026*

- **GPA:** 3.8 **| Awards:** Dean's List
- **Relevant Coursework:** Computational Fluid Dynamics, Aeronautical Propulsion, Heat Transfer, Introduction to Flight

SKILLS

Analysis: ANSYS Fluent (CFD), Python, MATLAB/Simulink, Thermal FEA, Tolerance Stackup, Material Selection
CAD and Manufacturing: SolidWorks - Modeling, Drawing, Simulation (CSWA Certified / Course Instructor), Familiar with Siemens NX, GD&T, DFM, Manufacturing Tooling Design, 3D Printing (FDM), Rapid Prototyping
Test and Metrology: Instron Testing, Keyence 3D Scanning, Drop and Thermal Testing, Dimensional Inspection, Root Cause Analysis
Languages: Spanish (Conversational), Turkish (Intermediate), Japanese (Intermediate)

PROFESSIONAL EXPERIENCE

Insulet Corporation Acton, MA
Mechanical Engineer - Equipment Design Co-op *July 2025 – December 2025*

- Designed and implemented high-precision automated manufacturing equipment using SolidWorks to improve product assembly accuracy and reliability for insulin pump production lines, incorporating GD&T and tolerance stackup to ensure ±0.05 mm alignment across assembly fixtures
- Performed root cause analysis and troubleshooting for critical equipment issues, diagnosing mechanical, pneumatic, and sensor-related failures on automated assembly stations to reduce unplanned downtime and improve OEE
- Developed 100+ part CAD models and 50+ production-ready engineering drawings with supporting thermal FEA simulations to validate equipment designs under operating conditions
- Collaborated with manufacturing engineers, technicians, and suppliers to execute design changes in a cGMP ISO 8 cleanroom
- Managed 5 equipment design projects from concept to installation in a regulated medical device environment, plus minor tooling and design-change requests

Thermacell Repellents Inc. Bedford, MA
Mechanical Engineer Co-op *July 2024 – December 2024*

- Owned the full design of a 500+ component impact test fixture – concept, tolerance stackup, material selection, prototyping, fabrication, and validation – used to qualify E65 rechargeable repellent units against various impact energies
- Supported the engineering team with hands-on testing including drop, temperature, and competitor benchmarking, as well as Keyence scanning of injection molded parts for dimensional inspection, analyzing results to inform design iterations
- Led FDM printing initiatives for the engineering department, managing in-house production of prototype and functional plastic parts to accelerate development timelines
- Created 100+ DFM compliant part and assembly drawings in SolidWorks per company drafting standards, coordinating with manufacturing vendors to ensure producibility

PROJECTS

Senior Capstone Design – Sponsored by Beta Technologies – Northeastern University *May 2026 – Dec 2026*

- Designing a combined flame-and-impact test rig to qualify eVTOL battery enclosure wall materials against thermal runaway containment; 5-person team advised by Prof. Juner Zhu
- Sizing the drop-weight impact architecture against sponsor load requirements (5-2,200 N, 300 N·s impulse) to bound carriage mass, drop height, and impactor geometry

Cessna 172S: Wing-Stabilizer Downwash Analysis – Computational Fluid Dynamics, Northeastern University *March 2026*

- Built a two-body 2D ANSYS Fluent model of a Cessna 172S NACA 2412 wing and NACA 0012 stabilizer at 10,000 ft cruise, deriving tail placement, incidence, and stabilizer MAC from POH weight-and-balance data and taper-ratio calculation
- Ran a five-case angle-of-attack sweep (0-8°) with SST k-ω closure, decomposing inlet velocity and lift/drag force vectors per attitude to simulate aircraft pitch without remeshing
- Quantified wing downwash effect on tail loading: stabilizer C_L fell from 0.323 to 0.186 across the sweep
- Correlated velocity and pressure contours with wake trajectory to confirm downstream deflection of wing wake onto stabilizer

ACTIVITIES

Northeastern American Society of Mechanical Engineers (ASME) Boston, MA
President *June 2026 – Present*
Vice President of Events, Social Chair, Company Liaison *Jan 2024 – May 2026*

- Lead a 17-member executive board overseeing all chapter operations, delegating responsibilities across specialized VP roles spanning event planning, corporate sponsorship and outreach to 100+ companies, a 50-student mentorship program, and communications
- Collaborate with the MIE department and partner organizations including AIChE, BMES, and IISE to align chapter programming with departmental goals and co-host career events
- Provide 200+ mechanical engineering students per semester with professional development through weekly industry speaker sessions, technical workshops, and the CSWA SolidWorks certification course